Practical 1

1. Title

Demonstration of Inline, Internal, and External JavaScript; Use of Console Methods; Creation of a Webpage Displaying User Information and a Welcome Message.

2. Software/Tools Required

- Visual Studio Code
- Google Chrome / Microsoft Edge
- HTML5
- JavaScript (ES6)

3. Theory

JavaScript is a lightweight, interpreted programming language that adds interactivity and dynamic behavior to web pages. It works together with HTML and CSS to create responsive web applications. JavaScript code can be included in an HTML document in three different ways:

- **Inline JavaScript:** Code is written directly inside an HTML element using event attributes such as `onclick`.
- **Internal JavaScript:** Code is written inside the `<script>` tag within the HTML document.
- **External JavaScript:** Code is written in a separate `.js` file and linked to the HTML document using the `src` attribute.

JavaScript also provides several console methods such as `console.log()`, `console.warn()`, `console.error()`, and `console.info()` which help developers debug programs and monitor execution. In this practical, a webpage is created to display user information and a welcome message using JavaScript. Console methods are also demonstrated to display different types of messages during execution.

4. Output

- A webpage displaying a welcome message.
- User information shown on the webpage.
- Console window displaying messages generated using various console methods.
- Navigation between webpages using JavaScript functionality.

Screenshots attached.

5. Result/Conclusion

The practical was successfully completed. Different methods of including JavaScript (inline, internal, and external) were demonstrated effectively. Console methods were used for debugging purposes, and a dynamic webpage displaying user information and a welcome message was successfully created, improving understanding of JavaScript integration with HTML.

Practical 2

1. Title

Implementation of JavaScript Variables (`var`, `let`, `const`), Template Literals, Destructuring, and Development of a Billing Calculator Using User Input.

2. Software/Tools Required

- Visual Studio Code
- Google Chrome / Microsoft Edge
- HTML5
- JavaScript (ES6)

3. Theory

JavaScript provides different ways to declare variables depending on their scope and mutability.

- **var:** Function-scoped variable that can be redeclared and updated.
- **let:** Block-scoped variable that can be updated but not redeclared within the same scope.
- **const:** Block-scoped constant whose value cannot be reassigned after initialization.

Modern JavaScript introduces several ES6 features that make coding simpler and more efficient.

- **Template Literals:** Allow embedding variables and expressions directly into strings using backticks (`` ` ``), making string formatting easier.
- **Destructuring:** Enables extraction of values from arrays or objects into separate variables in a concise manner.

In this practical, a billing calculator is developed where the user enters customer details, product details, quantity, and price per unit. JavaScript processes the input, calculates the total amount payable, and displays a formatted bill using template literals. The program demonstrates variable declarations, user input handling, arithmetic operations, and modern JavaScript features.

4. Output

- Customer name entered successfully.
- Product name, quantity, and price accepted as input.
- Total bill calculated automatically.
- Formatted billing statement displayed on the webpage.

Screenshots attached.

5. Result/Conclusion

The billing calculator was implemented successfully using JavaScript. The practical demonstrated the use of `var`, `let`, `const`, template literals, and destructuring concepts. The program correctly accepted user input, performed calculations, and displayed the generated bill in a user-friendly format, enhancing understanding of modern JavaScript programming.

Practical 3

1. Title

Implementation of Control Structures and Form Validation; Development of a Grading System Based on User-Entered Marks.

2. Software/Tools Required

- Visual Studio Code
- Google Chrome / Microsoft Edge
- HTML5
- JavaScript

3. Theory

Control structures are an essential part of JavaScript programming as they enable decision-making based on specific conditions. The commonly used conditional statements include:

- `if`
- `if...else`
- `else if`
- `switch`

These statements allow different blocks of code to execute depending on user input or program conditions.

Form validation is the process of checking whether user-entered data is correct and complete before processing it. Validation improves data accuracy and prevents invalid information from being submitted. Common validation techniques include checking for empty fields, ensuring marks are numeric, verifying that values fall within a valid range (0–100), and displaying appropriate error messages when invalid input is detected.

In this practical, the user enters marks through a form. JavaScript first validates the entered values and then uses conditional statements to determine the appropriate grade. Depending on the obtained marks, grades such as **A, B, C, D, or Fail** are displayed. Invalid or incomplete inputs are rejected with suitable warning messages.

4. Output

- Marks entered through the form.
- Validation performed before processing.
- Grade displayed based on the entered marks.
- Error messages shown for invalid or incomplete input.

Screenshots attached.

5. Result/Conclusion

The practical was successfully implemented using JavaScript control structures and form validation techniques. The program accurately validated user input, prevented incorrect data entry, and generated grades according to predefined conditions. This practical enhanced the understanding of decision-making statements, input validation, and interactive web application development using JavaScript.

Practical 4

1. Title

Implementation of JavaScript Function Types, Scope, and Closures; Exception Handling using Try-Catch; Development of a Palindrome Checker.

2. Software/Tools Required

- Visual Studio Code
- Google Chrome / Microsoft Edge
- HTML5
- CSS3
- JavaScript (ES6)

3. Theory

Functions are one of the fundamental building blocks of JavaScript and help in organizing code into reusable modules. JavaScript supports different types of functions, including function declarations, function expressions, arrow functions, and anonymous functions. Each type has its own use depending on the programming requirement.

JavaScript also provides different variable scopes, such as **global scope**, **function scope**, and **block scope**. Variables declared using `let` and `const` are block-scoped, whereas variables declared using `var` are function-scoped. Proper use of scope helps prevent unintended access or modification of variables and improves code maintainability.

A **closure** is a JavaScript feature where an inner function remembers and accesses variables from its outer function even after the outer function has finished execution. Closures are commonly used for data encapsulation, private variables, and creating reusable modules.

Exception handling is performed using the **try-catch** statement. The `try` block contains code that may generate an error, while the `catch` block handles the exception gracefully without terminating the program. This improves the reliability and robustness of web applications by displaying meaningful error messages to users.

In this practical, a **Palindrome Checker** is developed. The user enters a word or phrase, which is first validated using `try-catch`. The input is then normalized by removing spaces, punctuation, and converting all characters to lowercase. The cleaned string is compared with its reversed version to determine whether it is a palindrome. The application demonstrates the use of function expressions, closures, scoped helper functions, exception handling, event handling, and string manipulation to provide accurate results.

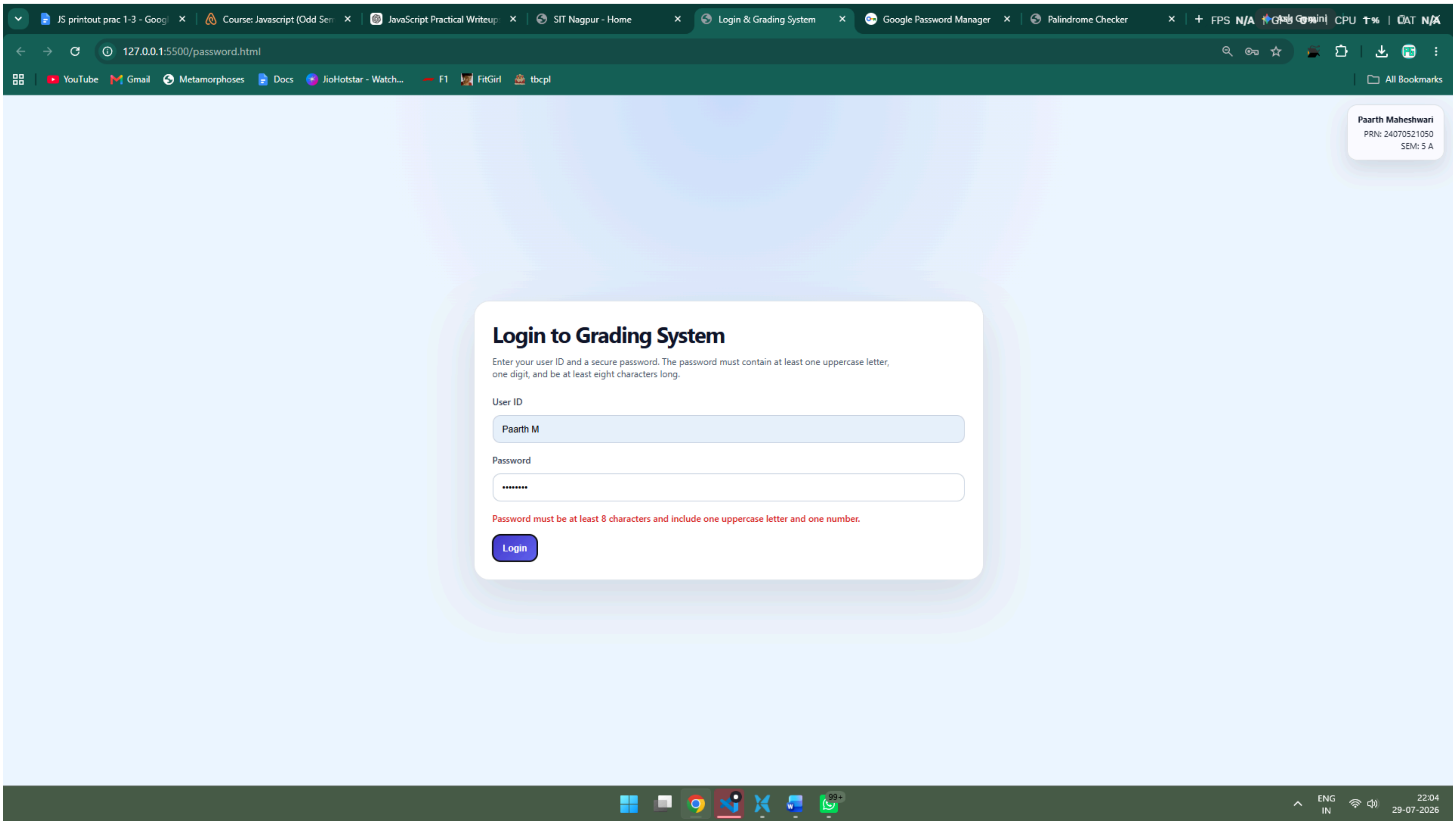
4. Output

- User enters a word or phrase in the input field.
- The program validates the entered data.
- Invalid or empty input displays an appropriate error message.
- Valid input is checked for palindrome.
- The webpage displays whether the entered text is a palindrome or not.
- A reset button clears the input field and output.

Screenshots attached.

5. Result/Conclusion

The practical was successfully implemented using JavaScript. Different function types, variable scope, closures, and exception handling using `try-catch` were effectively demonstrated. The palindrome checker accurately validated user input, handled errors gracefully, and correctly determined whether the entered word or phrase was a palindrome. This practical enhanced the understanding of modern JavaScript programming concepts and their application in developing interactive web applications.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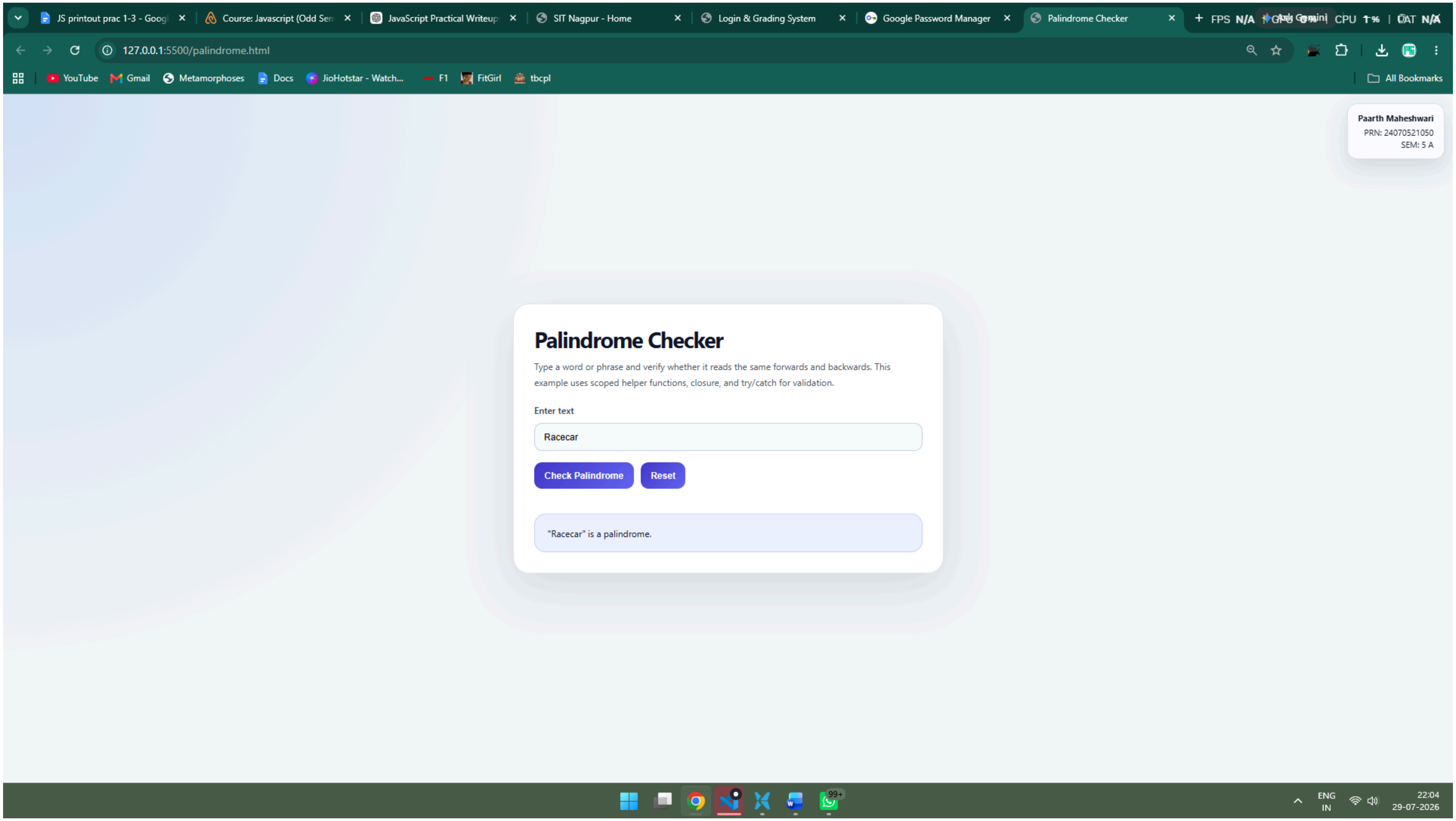
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Palindrome Checker
Type a word or phrase and verify whether it reads the same forwards and backwards. This example uses scoped helper functions, closure, and try/catch for validation.
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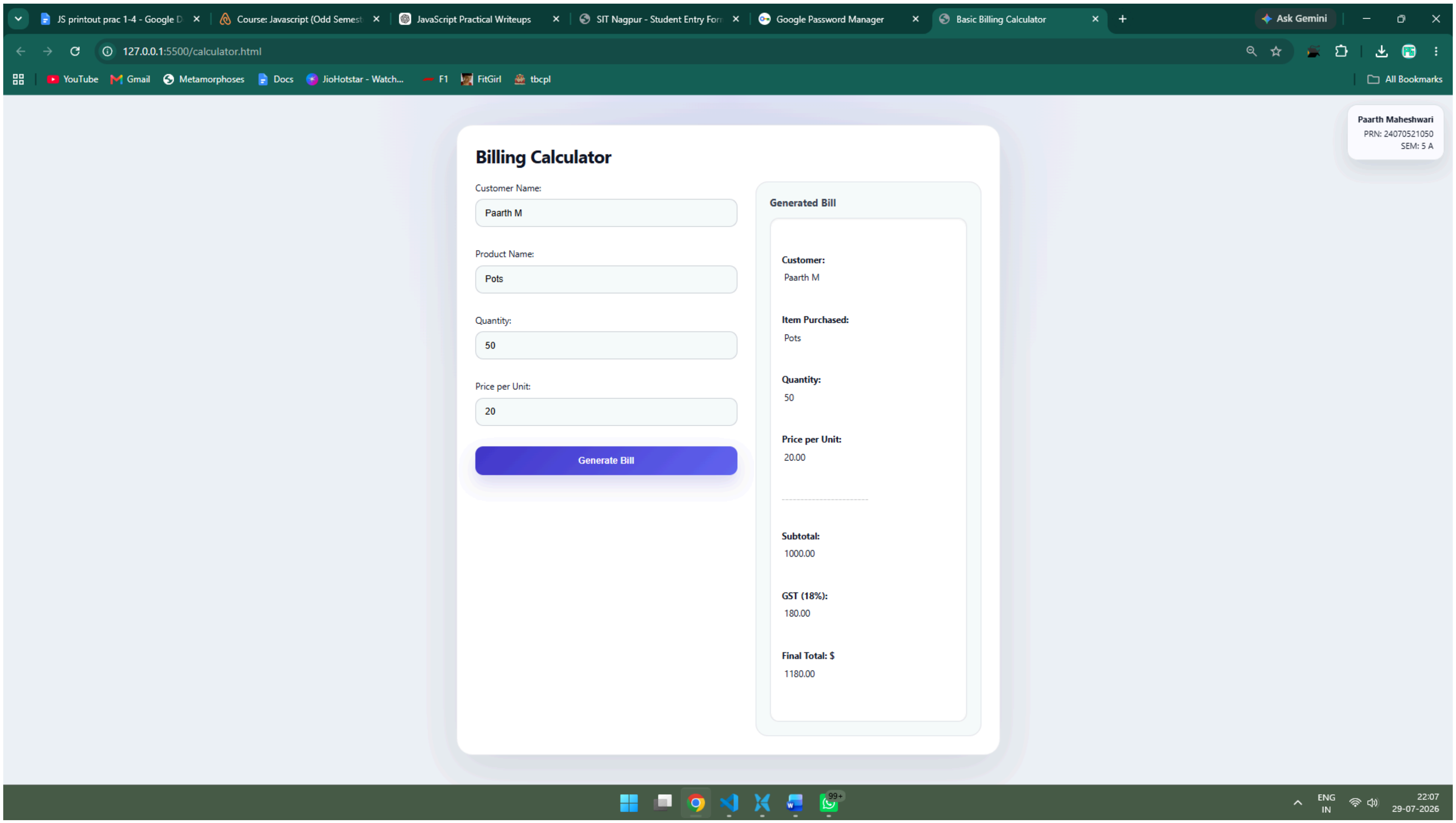

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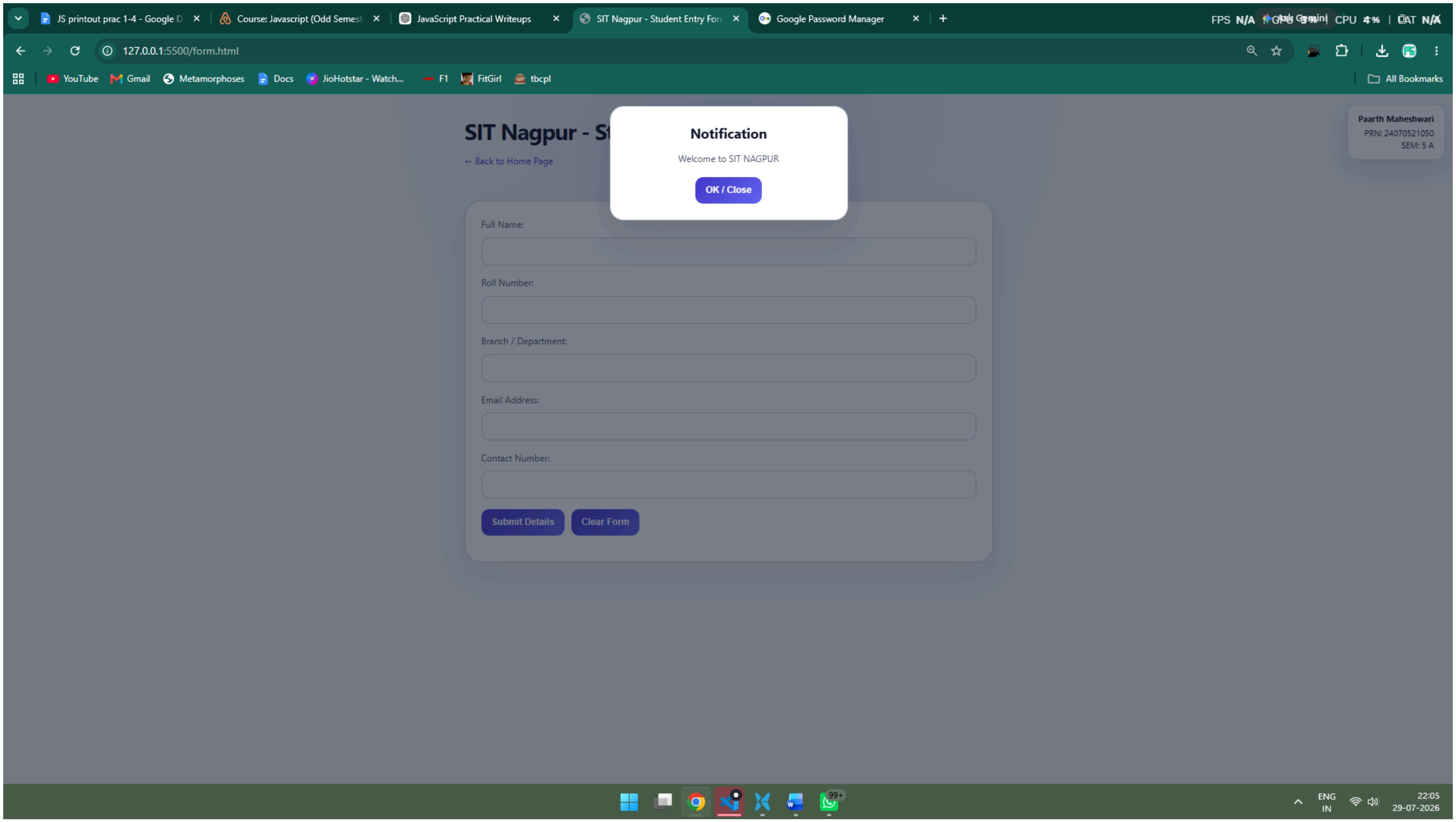
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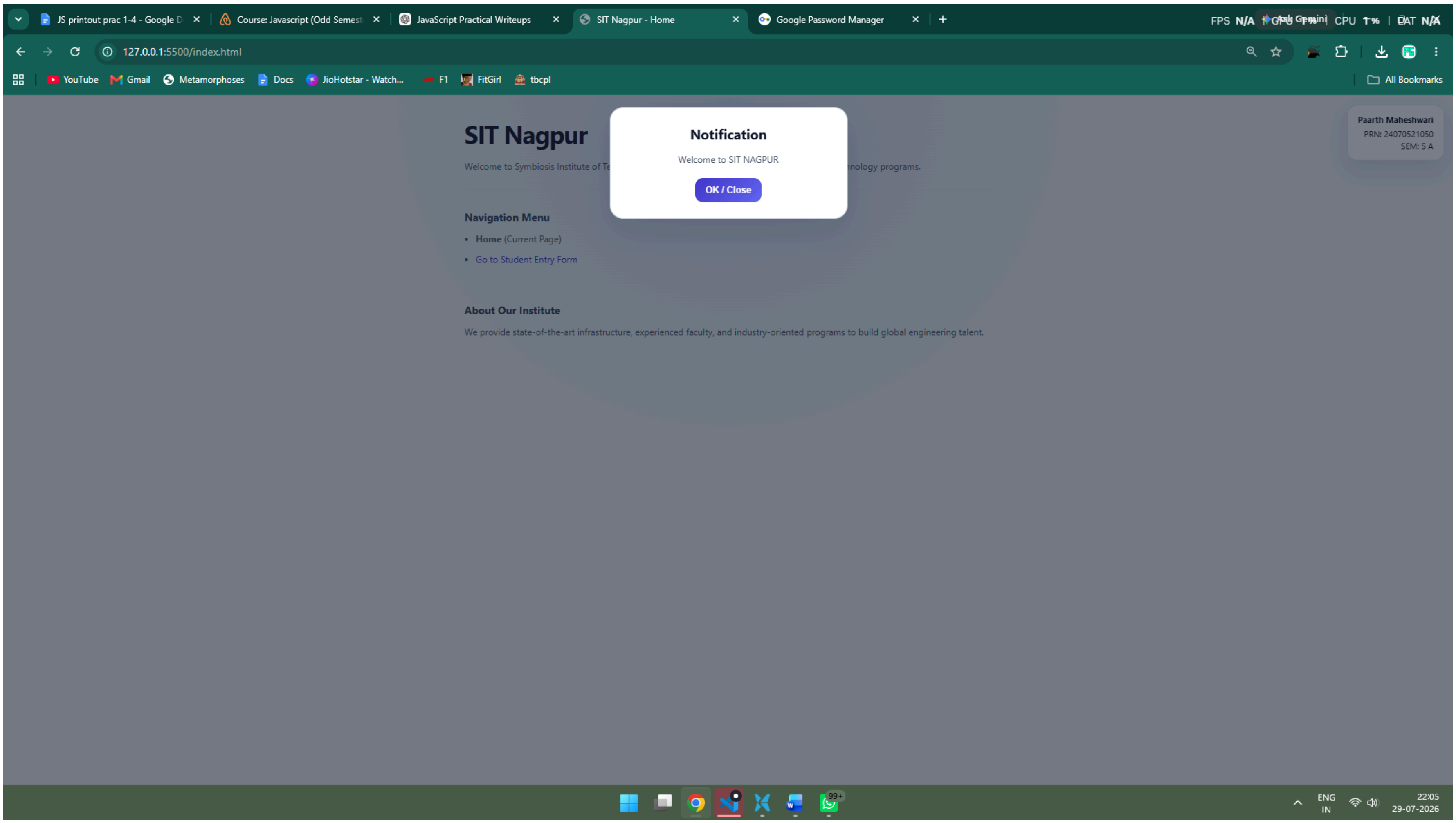
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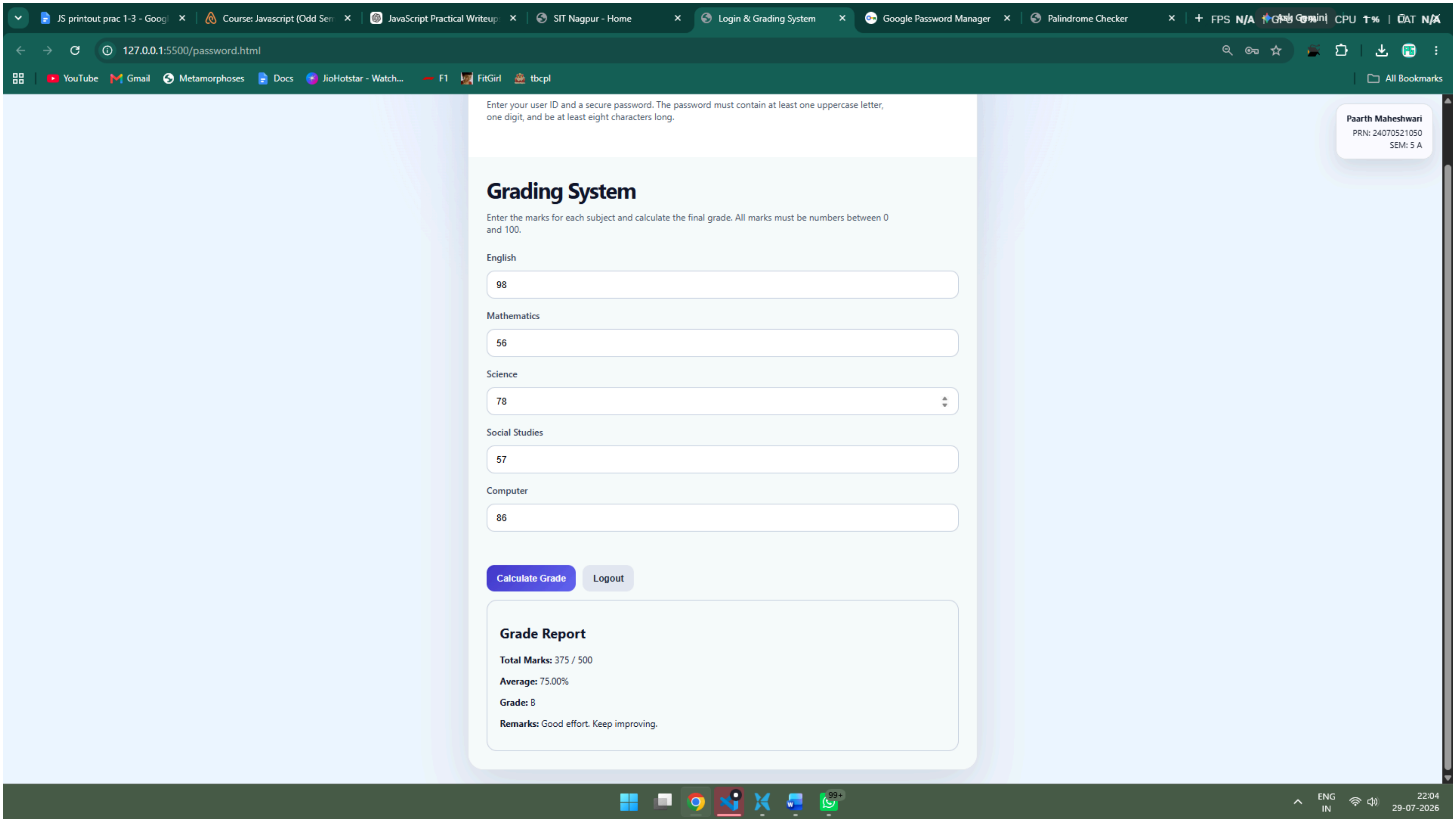
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Science
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Social Studies
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Average: 75.00%
Grade: B
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